

Group Project Report

Semantic Enrichment in Few-Shot Learning:
Bridging the Gap between Numeric IDs and Foundation Models

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<https://github.com/eliimirafiori/AML>



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1 Introduction

Few-Shot Learning (FSL) addresses a setting in which a model must classify images belonging to categories that were never observed during its main training phase, relying on only a handful of labelled examples provided at inference time. The specific configuration of this project is the canonical **5-way 5-shot** task: every evaluation episode supplies a *support set* of five novel classes, each illustrated by exactly five labelled images, and a *query set* of unlabelled images that must be assigned to one of those five classes. The model is expected to adapt to an entirely new set of categories in each episode, without ever updating its weights on the novel classes.

Leveraging Foundation Models. The natural way to attack such a problem is to exploit the prior knowledge of large pre-trained foundation models rather than to train a classifier from scratch. Models such as CLIP have been exposed to hundreds of millions of image-text pairs and have learned a joint embedding space in which visual and textual concepts are aligned; used as frozen feature extractors they transfer well to unseen categories, turning the problem from “learning to see” into the easier problem of “learning to compare”.

The Core Challenge: Anonymised Labels. The defining difficulty of our challenge, however, was a constraint on the labels themselves. The categories carried **no real names**: they were given purely as anonymous numerical identifiers such as Class 0, Class 1 and Class 2. This is a serious obstacle for a Vision-Language Model. CLIP achieves its zero-shot ability by comparing an image embedding against the embedding of a meaningful prompt such as “a photo of a kitchen”. When the only available prompt is “a photo of a [1]”, the text encoder receives a token with no semantic content, the alignment between language and vision is severed, and performance collapses to roughly 20% — which, on a five-class problem, is indistinguishable from chance.

Dataset Complexity. A second difficulty came from the images themselves: rather than isolated, centred objects, the dataset consisted of complex indoor and outdoor scenes (corridors, staircases, patios, living rooms) in which the discriminative signal is spread across background textures and global spatial layout.

Proposed Solution and Contributions. The literature suggests two complementary remedies. On the semantic side, recent multimodal models such as LLaVA can *reason* about an image in natural language, offering a way to manufacture the descriptive labels the dataset holds. On the representation side, a long line of work has shown that the quality of the frozen embedding matters more than any elaborate meta-learning algorithm, and that stronger or complementary backbones combined with a simple classifier are highly effective.

Following these ideas, we propose a pipeline that first restores semantic meaning through LLaVA and then progressively enriches the visual representation, culminating in a multi-backbone feature ensemble classified by a maximum-margin Support Vector Classifier. This system reached an accuracy of **96.93%** on the challenge with a standard deviation of only 0.38%.

1.1 Problem formalisation

We state the task formally. In each episode the model receives a labelled *support set*

$$S = \{(x_i, y_i)\}_{i=1}^{N \times K}, \quad y_i \in \{1, \dots, N\}, \quad (1)$$

containing N distinct novel classes (the *ways*) with exactly K labelled examples each (the *shots*); in our setting $N = K = 5$, so S holds 25 images. It also receives an unlabelled *query set* $Q = \{\tilde{x}_j\}_{j=1}^M$, and must assign each query image $\tilde{x}_j$ to one of the N support classes. Crucially, the N classes are drawn from a label space that is *disjoint* from the classes seen during pre-training, and no gradient update is performed on the support classes themselves.

The prediction rule depends on the stage of the pipeline, and our system uses two distinct mechanisms. In the *intermediate, prototype-based stages* (CLIP/CoCa with LLaVA text), a query $\tilde{x}$ is classified by nearest prototype under cosine similarity,

$$\hat{y}(\tilde{x}) = \arg \max_{c \in \{1, \dots, N\}} s(f(\tilde{x}), \mathbf{p}_c), \quad s(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad (2)$$

where $f(\cdot)$ is a frozen feature extractor and $\mathbf{p}_c$ is a class prototype built from the support information of class c . Because all embeddings in CLIP-style spaces are L_2 -normalised onto the unit hypersphere, the cosine is the natural similarity. In the *final stage*, however, we do not classify by similarity to a prototype: instead we form, for every support and query image, a concatenated descriptor $\mathbf{z}$ (defined in Section 3), fit a linear Support Vector Classifier

on the 25 labelled support descriptors of the episode, and assign each query by the classifier’s decision function. The two formulations share the same goal — separating the N classes in a frozen embedding space — but differ in how the decision boundary is obtained: a fixed prototype direction versus a learned maximum-margin hyperplane.

The anonymised-label constraint of this challenge attacks the prototype side directly: when the only available description of class c is a meaningless numeric token, the text prototype $\mathbf{p}_c^{\text{text}}$ carries no signal and the $\arg \max$ above degenerates to a random choice. Our method can be read as a systematic effort to make both the embedding f and the per-class support information as informative as possible, first by restoring semantics and then by replacing the fixed prototype with a learned margin.

2 Related Work

Our methodology draws on five distinct lines of prior research, each of which directly informed a specific architectural or design decision in our pipeline:

- **Good Embeddings for Few-Shot Learning:** Tian et al. [1] demonstrated that, contrary to the prevailing emphasis on elaborate meta-learning frameworks, the single most critical factor for FSL success is the quality of the learned representation. A strong feature embedding followed by a simple classifier consistently outperforms complex episodic training. This core insight underpins our entire strategy, steering our efforts toward producing rich, high-dimensional embeddings paired with deliberately lightweight classifiers.
- **Bridging the Semantic Gap:** Liu et al. [2] introduced LLaVA, a multimodal framework that couples a visual encoder with a large language model to reason about image content in natural language. We exploit precisely this capability to translate anonymous, non-semantic numeric labels into descriptive, meaningful textual identifiers.
- **Beyond Contrastive Matching:** Yu et al. [3] proposed CoCa, which augments the standard contrastive image–text objective with a generative captioning loss. Because the model is forced to explicitly *describe* images rather than merely match pairs, it captures finer visual nuances, making it a significantly more robust feature extractor than CLIP for detail-dependent scenes.
- **Combining Convolutions and Transformers:** Peng et al. [4] demonstrated that merging the local inductive bias of convolutional networks with the global receptive field of vision transformers yields representation spaces greater than the sum of their parts. This architectural synergy directly justifies our choice to concatenate ConvNeXt (for local micro-textures) with the Swin Transformer (for global spatial layout).
- **Maximising the Margin:** Chen et al. [5] showed that simple linear classifiers frequently outperform complex non-linear algorithms when the underlying features are sufficiently strong and high-dimensional. This provides the theoretical foundation for our final decision to deploy a linear Support Vector Classifier over the concatenated feature space.

3 Proposed Method

Our method was not designed in a single step; it emerged from a sequence of diagnoses, where at each stage we identified the precise reason the current system underperformed and introduced exactly the component needed to fix it. We first narrate this path and then describe each component in more detail.

3.1 The path we followed

As shown in figure 1, our final pipeline emerged from an iterative sequence of diagnoses: at each stage, we identified the precise bottleneck of the current architecture and introduced the exact component necessary to overcome it.

- **A Failing Zero-Shot Baseline:** We began by applying CLIP in a standard zero-shot fashion, building prompts directly from the anonymised class labels. This approach failed almost completely ($\sim 21\%$). The diagnosis was crucial: the failure was *not* visual — as CLIP’s image encoder produced robust representations — but strictly semantic, because the text descriptions it was compared against lacked any real-world meaning.

- **The Semantic Rescue:** To repair the broken language channel, we introduced LLaVA. By feeding it the support images, we obtained genuine descriptive captions, effectively translating an anonymous numeric identifier into a meaningful semantic label (e.g., “Tree” or “Statue”). Injecting these rich descriptions instead of empty numeric prompts into the text encoder restored the vision–language alignment, causing accuracy to leap from $\sim 21\%$ to over 80% — by far the largest single improvement of the project.
- **Using the Shots:** Even after the semantic rescue, the system still matched query images against text-derived prototypes, completely ignoring the visual information of the five support images per class. We corrected this by introducing a Linear Probe — a single linear layer trained directly on the frozen support features. This mechanism raised accuracy to approximately 91% and significantly reduced the episode-to-episode variance.
- **Seeing More Detail:** Standard CLIP is optimised primarily for global semantics, making it inherently weak at capturing the dense local details that distinguish complex indoor and outdoor scenes. Replacing it with CoCa, which is pre-trained to generate explicit captions and thus attends to finer structures, pushed our baseline performance well into the mid- 90% range.
- **Combining Strengths:** A single vision backbone eventually became an architectural bottleneck, as our classification tasks required simultaneously recognising tiny local textures and global spatial layouts. We resolved this by concatenating the features of three complementary backbones — CoCa, ConvNeXt, and the Swin Transformer. Classifying this unified 2560-dimensional vector with a linear Support Vector Classifier allowed us to reach our peak accuracy of 96.93% .

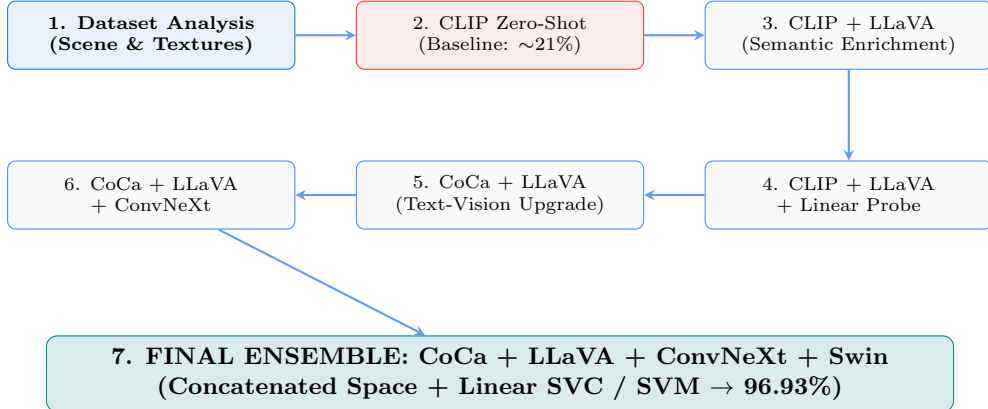


Figure 1: Incremental evolution of the proposed few-shot classification pipeline.

3.2 Components and their role

LLaVA for semantic enrichment. For each class c with support images $\{x_1, \dots, x_5\}$ we used LLaVA to obtain captions $\{t_1, \dots, t_k\}$, encoded each with the frozen text encoder ϕ_{text} , and averaged them into a normalised *text prototype*

$$\mathbf{p}_c^{\text{text}} = \text{norm} \left(\frac{1}{k} \sum_{i=1}^k \text{norm}(\phi_{\text{text}}(t_i)) \right), \quad (3)$$

where $\text{norm}(\cdot)$ denotes L_2 normalisation. Query images were then classified by cosine similarity to these prototypes. LLaVA converted a semantically empty problem into one the Vision-Language model could solve; it was run in 4-bit quantisation to fit a consumer GPU, and its captions were cached so they were produced only once.

The Linear Probe is the simplest trainable classifier: a single linear layer mapping a frozen feature vector to class logits under a cross-entropy loss. Its role was to let the system learn from the five support images per class. We kept it simple following the observation that with strong frozen embeddings a lightweight classifier is both sufficient and resistant to overfitting — a desirable property given only five examples per class.

CoCa augments CLIP’s contrastive objective with generative captioning, forcing the encoder to retain enough information to describe an image. For detail-rich scenes this yielded embeddings that encoded finer local structure, translating into a measurable gain when substituted for CLIP. The overall architectural paradigm of CoCa, which handles dual contrastive and generative tasks, is illustrated in fig. 2.

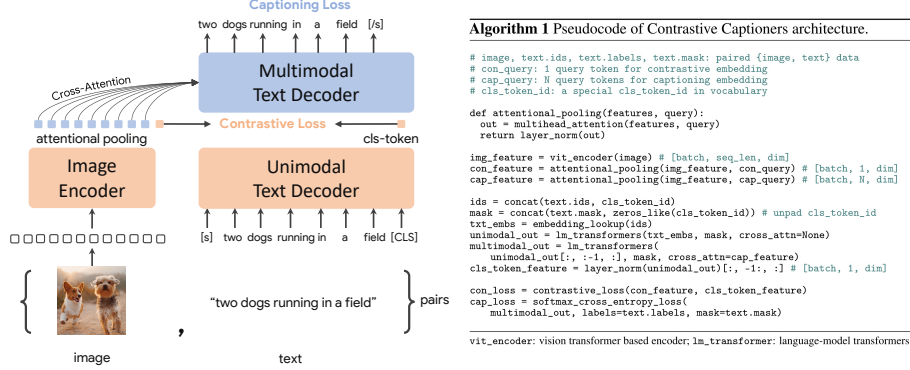


Figure 2: The CoCa foundational architecture combining contrastive and generative captioning objectives.

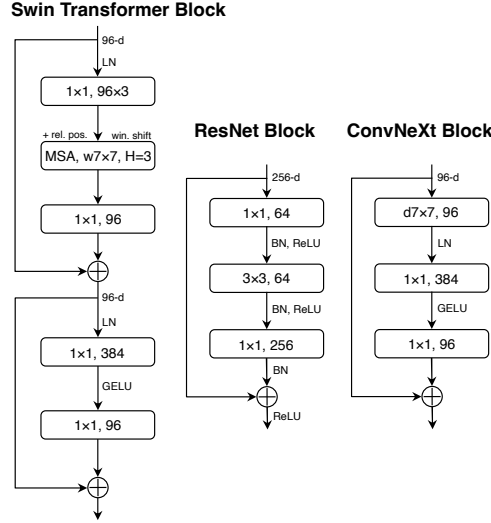


Figure 3: The macro-block structure and spatial downsampling pipeline of ConvNeXt.

ConvNeXt and Swin were chosen for their complementary inductive biases: ConvNeXt’s local receptive fields capture fine textures, while Swin’s shifted-window attention models global layout. The standard structural blocks and layers responsible for these features are referenced in fig. 3 for ConvNeXt and fig. 4 for the Swin Transformer. We combined them with CoCa by concatenation,

$$\mathbf{z} = [\mathbf{z}_{\text{CoCa}} \parallel \mathbf{z}_{\text{ConvNeXt}} \parallel \mathbf{z}_{\text{Swin}}] \in \mathbb{R}^{2560}, \quad (4)$$

preserving all information from each backbone and delegating feature selection to the final classifier. This was useful precisely because the three backbones make different errors.

The Linear Support Vector Classifier. The final stage abandons fixed prototypes in favour of a learned boundary. For each episode we build the concatenated descriptor $\mathbf{z}$ for all 25 support images and fit a linear Support Vector Classifier with regularisation constant $C = 0.50$ directly on these support points, $\text{clf.fit}(X_{\text{support}}, y_{\text{support}})$. Query images are then scored not by cosine similarity but by the classifier’s *decision function* — the signed geometric distance from the maximum-margin hyperplane. With strong, high-dimensional features a maximum-margin linear

classifier is inherently robust to overfitting, which is exactly the regime we faced with only 25 support points in a 2560-dimensional space. As a final refinement, because each episode contains a balanced number of queries per class, we convert the decision-function scores into an assignment problem and solve it with the Hungarian algorithm (`linear_sum_assignment`), which enforces the known class balance and yields the optimal global assignment of queries to classes rather than classifying each query in isolation. We additionally compared the Linear SVC against a Ridge Classifier and a Nearest-Centroid baseline on the same concatenated space; the SVC was both the most accurate and, decisively, the most stable.

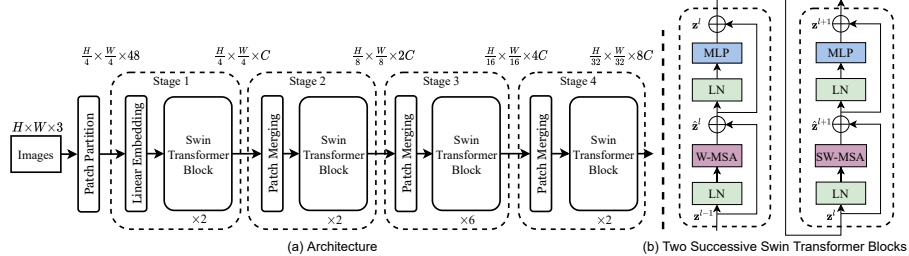


Figure 4: The shifted-window attention mechanism and hierarchical structure of the Swin Transformer.

4 Experiments

The incremental nature of our method lets us attribute a specific accuracy gain to each design decision. Table 1 summarises the full progression, reporting both the mean accuracy and the standard deviation across evaluation episodes.

Table 1: Ablation study across the incremental pipeline.

Phase	Configuration	Acc.	Std	Effect
Step 1	Standard CLIP (Zero-Shot)	21.40%	$\pm 8.60\%$	Fails on numeric labels
Step 1b	CLIP + LLaVA (Text-Enriched)	82.80%	$\pm 10.81\%$	Semantic meaning restored
Step 2	CLIP + LLaVA + Linear Probe	91.00%	$\pm 7.58\%$	Support set learned
Step 2b	CoCa + Probe Tuning	93.80%	N/A	Finer visual detail
Step 3	CoCa + ConvNeXt	94.80%	$\pm 6.46\%$	Micro-textures captured
Final	Ensemble (2560-d) + Linear SVC	96.93%	$\pm 0.38\%$	Max margin, no overfit

Local Validation vs. Kaggle Submission. It is important to clarify the evaluation protocol behind the results reported in Table 1. Due to the submission limits of the competition, these local metrics were computed by simulating thousands of pseudo-episodes extracted entirely from the 5000 available training images, rather than directly using the official hidden test episodes. Throughout our testing cycles, we observed a consistent generalization gap between our local cross-validation and the Kaggle platform: on average, performance dropped by approximately 2% when moving from local episodes to the online test sets. Consequently, while our final multi-backbone ensemble achieved a peak accuracy of 96.93% in our local environment, it stabilized at a final score of 94.60% on the official Kaggle leaderboard.

What worked: the semantic rescue. The largest single jump — from 21.40% to 82.80%, more than sixty percentage points — was produced entirely by restoring language through LLaVA, with no change to the vision backbone. This confirmed our hypothesis that on an anonymised dataset the dominant bottleneck is semantic rather than visual. Once meaningful text was supplied, the latent capability of the Vision-Language model was unlocked.

What worked: learning from the shots. Adding the Linear Probe lifted accuracy to 91.00% and reduced the standard deviation from 10.81% to 7.58%, showing that explicitly training on the support features both improved and stabilised the model that had previously ignored the visual examples.

What worked: stronger and more diverse vision. Replacing CLIP with CoCa (93.80%) and then adding

ConvNeXt (94.80%) produced steady, complementary gains, validating that generative captioning and local convolutional detail capture information the contrastive global features miss.

The decisive role of the ensemble and SVM. The final transition to the Linear SVC on the concatenated 2560-dimensional space did more than maximise accuracy at 96.93%: it collapsed the standard deviation from roughly 10% in the early phases to a mere 0.38%. This stabilisation is as significant as the headline accuracy, since it shows the high-dimensional ensemble generalises consistently across episodes rather than succeeding only on favourable ones.

4.1 A bias–variance reading of the results

The progression in Table 1 admits a natural *qualitative* interpretation through the lens of the bias–variance trade-off. We stress that this is an interpretive reading rather than a formal decomposition: the classical identity

$$\mathbb{E}[(\hat{y} - y)^2] = \underbrace{(\mathbb{E}[\hat{y}] - y)^2}_{\text{bias}^2} + \underbrace{\mathbb{E}[(\hat{y} - \mathbb{E}[\hat{y}])^2]}_{\text{variance}} + \sigma^2 \quad (5)$$

is stated for squared-error regression, whereas our task is classification; nevertheless it offers a useful conceptual frame. We treat the mean accuracy as an empirical proxy for (the complement of) bias — how systematically correct the classifier is — and the standard deviation across episodes as a proxy for variance — how much the result fluctuates with the random draw of support and query images in each episode.

Episodic Evolution through Bias and Variance. Read in this light, the incremental pipeline tells a coherent story. The initial zero-shot baseline behaves like a textbook *high-bias* system: constrained by semantically meaningless numerical prototypes, it is systematically wrong, and its 21.40% accuracy sits directly at the random floor. The subsequent semantic rescue via LLaVA sharply reduces this systemic bias, as the resulting prototypes finally point in meaningful directions within the joint embedding space. However, the episodic variance remains remarkably high ($\pm 10.81\%$) because a single noisy caption or an unrepresentative support image can still drastically swing the overall alignment of an entire episode.

To counteract this instability, each subsequent design decision acts chiefly as a variance-reduction mechanism. The introduction of the Linear Probe, by explicitly learning from the frozen support features, tightens the classification boundaries and drops the standard deviation to 7.58%. Next, the inclusion of complementary vision backbones introduces decorrelated architectural evidence whose idiosyncratic representation errors naturally tend to cancel out under feature concatenation.

This engineering progression culminates in the transition to the linear Support Vector Classifier, which brings a dramatic collapse of the standard deviation down to a mere 0.38%. This behavior matches core insights from few-shot literature: with only 25 support points in a high-dimensional 2560-dimensional space, a flexible classifier would possess excessive capacity, overfitting to noise and inflating variance. The linear maximum-margin hyperplane instead acts as an implicit capacity-control mechanism, trading a minimal amount of flexibility for massive stability across highly diverse evaluation episodes.

Analysis of Discarded Methodologies. Earlier in the project, we explored alternative avenues that were ultimately discarded due to poor performance. Purely visual prototypes built from the five support images alone proved noisy and unstable, confirming that five visual samples are insufficient to characterize an entire scene category robustly, whereas the language-guided text channel turned out to be far more stable.

Similarly, we experimented with test-time augmentation (TTA), including horizontal flips and crops, but these transformations slightly *degraded* final performance. Because many indoor and outdoor architectural layouts lack left–right symmetry, flipping introduced misleading structural variations rather than useful geometric invariances. These negative results reinforced our central conclusion that, on this specific dataset, reconstructing explicit semantic information is far more impactful than any form of visual data augmentation.

5 Conclusion

The story of this project is one of diagnosing the right problem before attempting to solve it. Our initial failure was easy to misread as a weakness of the vision model, but it was in fact a consequence of the missing semantic channel. Recognising this allowed us to deploy LLaVA to reconstruct the natural-language labels the dataset withheld, which produced the decisive improvement. From there, every further gain came from a disciplined sequence of steps, each addressing a concrete shortcoming of the one before: exploiting the support images with a Linear Probe, sharpening the visual representation with CoCa, and finally fusing complementary backbones into a single descriptor classified by a maximum-margin SVM.

Future Work: Targeted Segmentation and Augmentation. To counteract the data scarcity inherent to the few-shot regime, a promising direction involves the integration of foundational segmentation networks, such as the Segment Anything Model (SAM). By automatically isolating and generating precise masks exclusively for images that exhibit a clear, prominent foreground subject, we can implement a highly controlled data augmentation strategy. This would expand the available dataset while allowing Vision Transformers (ViTs) to maximize their intrinsic capacity for recognizing well-defined visual structures, free from background noise.

Future Work: Alternative Supervised Classifiers. While the linear Support Vector Machine provided exceptional stability and a robust global optimum for our final space, the classification layer remains open to alternative paradigms. Future efforts will systematically evaluate different downstream supervisors — specifically Ridge Classifiers and Nearest-Centroid baselines — to explore whether alternative regularization or geometric prototypes can offer comparable maximum-margin benefits on the concatenated representations.

Future Work: Latency Reduction and Heuristic Baselines. A final critical axis for development is operational efficiency and the reduction of overall analysis time. This will be pursued by investigating lighter, distilled foundation backbones that minimize computational overhead. Concurrently, we aim to benchmark purely heuristic or closed-form mathematical feature-selection methods against our multi-model ensemble. Testing these mathematical alternatives will determine if lean, deterministic algorithms can perform on par with complex deep learning pipelines while offering near-instantaneous execution times.

6 Individual Contribution

- **Elia Mirafiori:** He focused on the theoretical side, the core design of the architecture, and the general coordination of the project. He set up the main guidelines for our pipeline, led the first phase of the project, and managed the testing cycles and workload distribution. He also wrote and structured the final report together with Vittorio.
- **Filippo Segalla:** He took care of the compute infrastructure, backend development, and hardware setup. He provided the heavy-duty computing power needed for the resource-intensive LLaVA tasks. On top of that, he built the feature-extraction pipelines and integrated the multi-backbone ensemble with the final linear SVM classifier.
- **Vittorio Pettene:** He was responsible for the evaluation, benchmarking, and validation phases. He handled the main LLaVA testing loops, ran the ablation studies that proved that semantic enrichment works better than standard zero-shot architectures, and managed the high-performance environments. He also worked closely with Elia on writing and compiling this final report.
- **Tommaso Delucchi:** He played a key role in the testing, architecture, and presentation phases. In the beginning, he ran the initial tests with CLIP under Elia’s guidance. He also came up with the idea of condensing LLaVA’s long captions into a single descriptive keyword to make the text prototypes sharper. Later, he implemented ConvNeXt and Swin Transformer into the pipeline and tested them. Finally, he led the creation of the presentation slides and helped writing this report.

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